Student Grade Analysis

Data Mining and Predictive Modeling of Academic Performance

Project Report

Tools: Python · pandas · SciPy · statsmodels · scikit-learn · SHAP

Dataset: student_data.csv (395 students, 33 attributes)

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1. Executive Summary

This project investigates which social, family, behavioural and academic factors are associated with secondary-school students' grades, and builds regression models to predict them. It combines exploratory analysis, hypothesis testing (t-test and ANOVA), four regression algorithms and SHAP-based model interpretation on a dataset of 395 students.

Key findings

- **Internet access:** students with home internet had a higher average grade (10.84 vs 9.88; $p = 0.0415$).
- **Parental education:** mother's education is significantly associated with grades (ANOVA $p = 0.013$); father's education is not ($p = 0.499$).
- **Prediction:** without the grade columns as inputs, the best model (Random Forest) reaches $R^2 = 0.26$ and RMSE = 3.30 on a 0–20 scale.
- **Drivers:** SHAP ranks past failures and absences as the most influential features, followed by health, school support and parental education.
- **Consistency note:** a few headline claims in the notebook and README ($R^2 = 0.85$, $p < 0.001$, alcohol effect) are not supported by the computed outputs. Section 8 details this.

2. Project Overview

2.1 Objectives

- Analyse how social, behavioural and academic factors relate to student performance.
- Apply regression models to predict students' grades.
- Interpret results with statistical tests and SHAP feature importance.
- Provide practical recommendations for schools.

2.2 Dataset

The data come from the Kaggle "Student Performance" dataset: 395 students and 33 attributes with no missing values and no duplicate rows. Attributes cover demographics (sex, age, address), family background (parents' education and jobs, family size), study habits (study time, failures, extra paid classes), lifestyle (going out, workday and weekend alcohol use, health) and three grades: G1, G2 and the final grade G3, all on a 0–20 scale.

2.3 Tools and Project Files

Item	Description
student_analysis.ipynb	Full analysis notebook: cleaning, EDA, statistics, modeling, SHAP
student_data.csv	Kaggle dataset (395 × 33)
analysis_report.txt	Saved model metrics and statistical test output
project-report.pdf, .docx	Course reports (Persian) summarising the results

Item	Description
Libraries	pandas, NumPy, SciPy, statsmodels, scikit-learn, SHAP, Matplotlib, Seaborn

3. Data Preparation and Feature Engineering

The notebook validates the data, treats outliers and creates new features before modeling.

- **Validation:** duplicates were checked (none found) and grades clipped to the 0–20 range.
- **Outliers:** winsorization at the 5th and 95th percentiles was applied to age, absences, travel time, family relationship, free time, going out, alcohol use, health and the three grades.
- **Encoding:** 17 categorical columns were one-hot encoded (first level dropped), giving a numeric feature matrix.
- **Scaling:** features were standardised with StandardScaler before modeling.

Four features were engineered:

Feature	Formula	Purpose
average_grade	mean(G1, G2, G3)	Prediction target and overall performance score
grade_progression	G3 − G1	Improvement during the year
parent_edu	Medu + Fedu	Combined parental education
study_effort	studytime × (5 − failures)	Study time adjusted for past failures

G1, G2, G3, average_grade and grade_progression were removed from the model inputs so that the models predict performance from background and behavioural factors rather than from earlier grades.

4. Exploratory Data Analysis

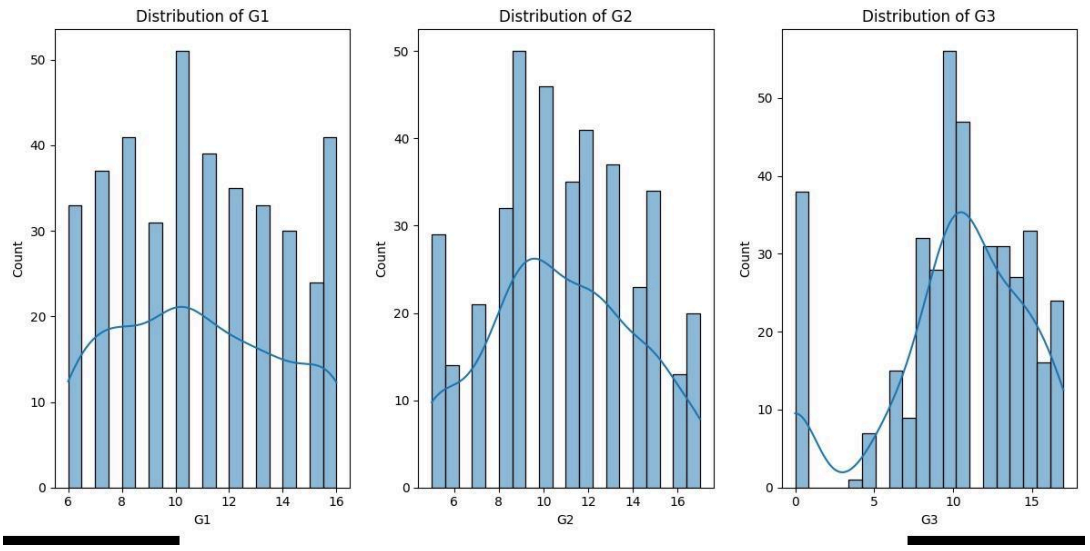


Figure 1: Distributions of G1, G2 and G3

G1 and G2 are spread fairly evenly between about 6 and 17. G3 is concentrated around 9–15 but has a separate spike at 0 (38 students, likely dropouts or missing final exams). This zero group makes the final grade noticeably less regular than the earlier ones.

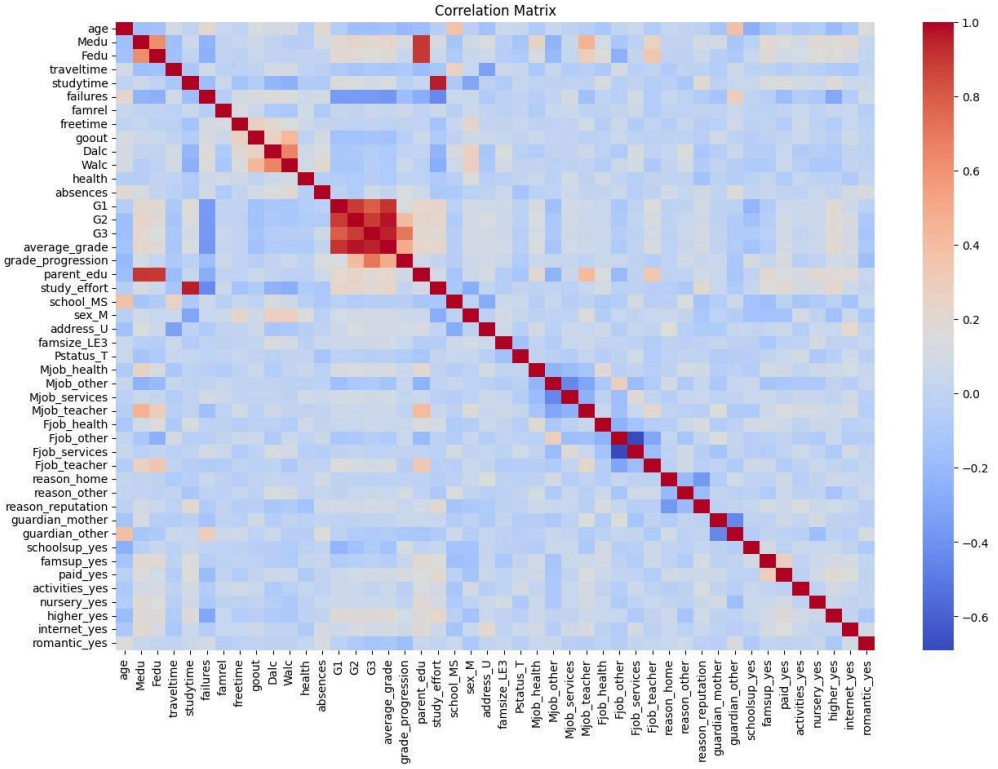


Figure 2: Correlation matrix of all encoded features

The three grades are strongly inter-correlated, and the engineered features (average_grade, parent_edu, study_effort) correlate with their source variables, as expected. Past failures show the clearest negative relationship with grades ($r \approx -0.38$ with average grade), while parental education shows a modest positive one. Mother's and father's education are themselves highly correlated.

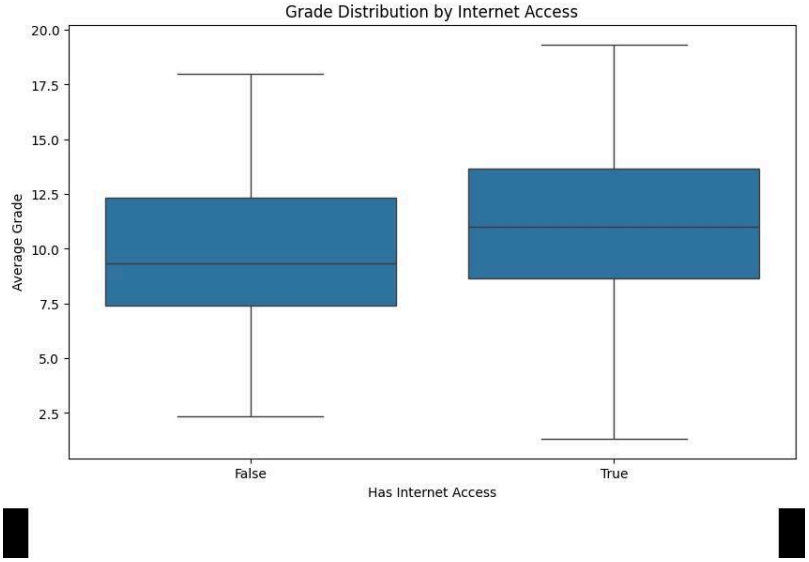


Figure 3: Average grade by internet access

Students with internet access have a higher median and a higher upper quartile than those without, although the two distributions overlap considerably.

5. Statistical Analysis

5.1 Internet Access (independent t-test)

Average grade was compared between students with internet access (n = 329) and without (n = 66).

Group	Mean average grade	Result
With internet (n = 329)	10.84	t = 2.05
Without internet (n = 66)	9.88	p = 0.0415 (significant at 0.05)

The difference of roughly one grade point is statistically significant but only modestly so; it would not remain significant at the 0.01 level.

5.2 Parental Education (two-factor ANOVA)

Factor	Sum sq.	df	F	p-value	Verdict
Mother's education	165.78	4	3.20	0.0132	Significant
Father's education	43.61	4	0.84	0.4990	Not significant
Residual	4997.24	386	—	—	—

Average grade rises with mother's education: about 9.25 for level 1 (n = 59), 10.24 for level 2, 10.52 for level 3 and 11.75 for level 4 (n = 131). Because the two parents' education levels are correlated, the father's effect may be absorbed by the mother's in this model. The effect size is also small: parental education explains only a few percent of the variance in grades.

6. Predictive Modeling

Four regression models predicted average_grade from the scaled feature set, using an 80/20 train–test split (random_state = 42). Models were evaluated by RMSE and R^2 on the test set.

Model	RMSE	R^2	Rank
Linear Regression	3.41	0.22	3 (tie)
Ridge (α = 0.5)	3.41	0.22	3 (tie)
Gradient Boosting	3.34	0.25	2
Random Forest (100 trees)	3.30	0.26	1

All four models perform similarly, and the non-linear ensemble methods give only a small gain over linear regression. An R^2 of about 0.26 means the models explain roughly a quarter of the variation in grades; background factors alone are only partly predictive. Random Forest was selected for interpretation.

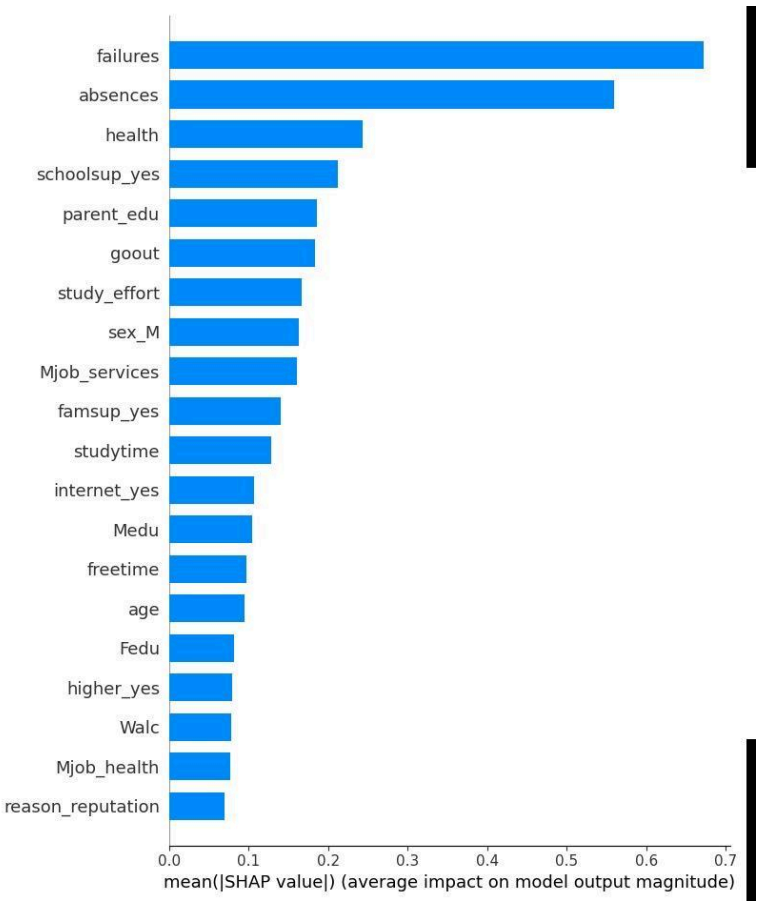


Figure 4: SHAP feature importance for the Random Forest model

The SHAP bar chart shows mean absolute impact on predicted grade. Failures (about 0.67) and absences (about 0.56) are clearly the largest contributors, followed by health, school support, parental education, going out, study effort and sex. Internet access and mother's education appear in the middle of the ranking, and weekend alcohol use (Walc) ranks near the bottom of the displayed features.

7. Insights and Recommendations

Insight	Supporting evidence
Past failures and absences are the strongest signals	Top two SHAP features; failures correlate negatively with grades
Internet access is linked to higher grades	Mean 10.84 vs 9.88, p = 0.0415
Mother's education matters more than father's	ANOVA p = 0.013 vs 0.499
Models give moderate predictive power	Best R^2 = 0.26, RMSE = 3.30

Recommendations

- **Early-warning system:** flag students with previous failures and rising absences for counselling and mentoring.
- **Internet access programs:** provide school-based or subsidised access for students without it, while recognising this may be a proxy for household resources.
- **Parent engagement workshops:** support families, especially where parental education is lower, in helping with study.
- **Wellbeing awareness:** run health and alcohol-awareness activities, treating them as supportive measures rather than proven grade boosters (see Section 8).

8. Limitations and Result Consistency

While preparing this report, the reported claims were compared with the notebook's computed outputs. Several statements should be corrected or qualified before the work is cited.

Claim in notebook / README	Computed result	Suggested action
Random Forest $R^2 = 0.85$	$R^2 = 0.26$ in analysis_report.txt; 0.85 is a hard-coded text line	Report 0.26, or document the tuning that gives 0.85
Internet effect $p < 0.001$	$p = 0.0415$	Report $p = 0.0415$
Weekend alcohol harms grades	Not tested; correlation with grades ≈ -0.09, low SHAP rank	Present as a weak, untested association
65% / 35% train–test split (README)	Code uses 80% / 20%	Align README with code

Further limitations:

- All findings are associations, not causal effects; for example, internet access may reflect household income.
- Winsorization and the average_grade target blur the zero-grade group, which behaves differently from other students.
- Evaluation uses a single train–test split; cross-validation would give a more reliable estimate.
- Several models and tests were run without multiple-comparison correction.
- SHAP bars show importance but not direction; the colour/direction interpretations in the Persian report cannot be read from this bar chart.

9. Conclusion

The analysis shows that past failures, absences, home internet access and mother's education are all linked to student grades, with failures and absences being the most influential in the Random Forest model. Predictive accuracy from non-grade factors is moderate ($R^2 \approx 0.26$), which is useful for identifying students who may need support but not for precise grade forecasting. Future work should add cross-validation and hyperparameter tuning with reported results, test the alcohol hypothesis directly, and consider models that include earlier grades (G1, G2) when the goal is accurate prediction rather than explanation.